

MATH 465 – Homework 5 Solutions

Due October 8, 2026

Name: _____

Acknowledgment. I consulted the homework prompt and the course lecture notes. No students were consulted. Generative AI assistance was used in preparing this write-up.

1. Finite differences and interpolation

Define

$$q(t) = p(2t).$$

Then q has degree at most 4, and the given values of p give the following forward-difference table:

t	0	1	2	3	4
$q(t)$	1	4	13	40	109
$\Delta q(t)$	3	9	27	69	
$\Delta^2 q(t)$	6	18	42		
$\Delta^3 q(t)$	12	24			
$\Delta^4 q(t)$	12				

The fourth difference is constant because q has degree at most 4.

(a) Values at 10 and -2

To extend the table to $t = 5$, first extend the successive differences:

$$\Delta^3 q(4) = \Delta^3 q(3) + \Delta^4 q(3) = 24 + 12 = 36,$$

$$\Delta^2 q(4) = \Delta^2 q(3) + \Delta^3 q(4) = 42 + 36 = 78,$$

$$\Delta q(4) = \Delta q(3) + \Delta^2 q(4) = 69 + 78 = 147.$$

Therefore

$$q(5) = q(4) + \Delta q(4) = 109 + 147 = 256,$$

and hence

$$\boxed{p(10) = 256}.$$

Newton's forward interpolation formula gives

$$q(t) = q(0) + \Delta q(0) \binom{t}{1} + \Delta^2 q(0) \binom{t}{2} + \Delta^3 q(0) \binom{t}{3} + \Delta^4 q(0) \binom{t}{4},$$

so in this case

$$q(t) = 1 + 3 \binom{t}{1} + 6 \binom{t}{2} + 12 \binom{t}{3} + 12 \binom{t}{4}.$$

At $t = -1$, the generalized binomial identity $\binom{-1}{k} = (-1)^k$ gives

$$q(-1) = 1 - 3 + 6 - 12 + 12 = 4.$$

Since $p(-2) = q(-1)$,

$$\boxed{p(-2) = 4}.$$

(b) Formula and uniqueness

Substituting $t = x/2$ into the interpolation formula yields

$$p(x) = 1 + 3\binom{x/2}{1} + 6\binom{x/2}{2} + 12\binom{x/2}{3} + 12\binom{x/2}{4}.$$

This polynomial is uniquely determined by the five data values. Indeed, if p_1 and p_2 both have degree at most 4 and satisfy the given conditions, then $p_1 - p_2$ has the five distinct roots 0, 2, 4, 6, 8. A nonzero polynomial of degree at most 4 cannot have five distinct roots, so $p_1 - p_2$ is identically zero.

2. Integer-valued polynomials

(a) Integer values at every integer

Let $\Delta f(x) = f(x+1) - f(x)$. The Newton expansion for a polynomial of degree at most d is

$$p(x) = \sum_{k=0}^d \Delta^k p(0) \binom{x}{k}, \quad \binom{x}{k} = \frac{x(x-1)\cdots(x-k+1)}{k!}.$$

To justify this expansion, note that $\Delta \binom{x}{k} = \binom{x}{k-1}$, with $\binom{x}{0} = 1$, so the successive differences of the right-hand side at $x = 0$ are exactly $\Delta^k p(0)$. The two sides are polynomials of degree at most d with the same values at $0, 1, \dots, d$, and therefore they are equal.

Because $p(0), \dots, p(d)$ are integers, every $\Delta^k p(0)$ is an integer: each finite difference is an integer linear combination of these given values. For a nonnegative integer m , $\binom{m}{k}$ is an integer in the usual sense. For a negative integer m ,

$$\binom{m}{k} = (-1)^k \binom{-m+k-1}{k},$$

which is also an integer. Thus every term in the Newton expansion is an integer when $x = m \in \mathbb{Z}$, and consequently

$$p(m) \in \mathbb{Z} \quad \text{for every } m \in \mathbb{Z}.$$

(b) Ordinary coefficients need not be integers

No. For example, take $d = 2$ and

$$p(x) = \binom{x}{2} = \frac{x^2 - x}{2}.$$

Then

$$p(0) = 0, \quad p(1) = 0, \quad p(2) = 1$$

are integers, and part (a) shows that $p(m)$ is an integer for every integer m . However, the coefficients of x^2 and x in the ordinary power basis are $1/2$ and $-1/2$, respectively. Hence integer-valuedness does not imply integer coefficients in the power basis.

3. A Stirling-number summation identity

(a) Proof of the identity

Using the permitted identity

$$j^m = \sum_{k=0}^m S(m, k)(j)_k, \quad (j)_k = j(j-1) \cdots (j-k+1),$$

we obtain

$$\begin{aligned} \sum_{j=0}^{N-1} j^m &= \sum_{j=0}^{N-1} \sum_{k=0}^m S(m, k)(j)_k \\ &= \sum_{k=0}^m S(m, k) \sum_{j=0}^{N-1} k! \binom{j}{k} \\ &= \sum_{k=0}^m k! S(m, k) \sum_{j=0}^{N-1} \binom{j}{k}. \end{aligned}$$

The hockey-stick identity gives

$$\sum_{j=0}^{N-1} \binom{j}{k} = \binom{N}{k+1}.$$

This remains true for $k = 0$, and when $N = 0$ both sides of the desired identity are empty sums or zero binomial coefficients. Therefore

$$\boxed{\sum_{j=0}^{N-1} j^m = \sum_{k=0}^m k! S(m, k) \binom{N}{k+1}.}$$

(b) The fourth-power sum

The relevant Stirling numbers of the second kind are

$$S(4, 0) = 0, \quad S(4, 1) = 1, \quad S(4, 2) = 7, \quad S(4, 3) = 6, \quad S(4, 4) = 1.$$

Multiplying by the corresponding factorials gives

k	0	1	2	3	4
$k!S(4, k)$	0	1	14	36	24

and hence

$$\boxed{\sum_{j=0}^{N-1} j^4 = \binom{N}{2} + 14 \binom{N}{3} + 36 \binom{N}{4} + 24 \binom{N}{5}.}$$

4. A nonhomogeneous linear recurrence

(a) A resonant trial sequence cannot work

If $h_n = A2^n$, then the left-hand side of the recurrence is

$$\begin{aligned} h_n - 3h_{n-1} + 2h_{n-2} &= A2^{n-2}(4 - 6 + 2) \\ &= 0. \end{aligned}$$

The right-hand side is $2^n \neq 0$. Thus no sequence of the form $h_n = A2^n$, with A constant, satisfies the recurrence.

(b) Closed formula

The homogeneous recurrence has characteristic polynomial

$$\lambda^2 - 3\lambda + 2 = (\lambda - 1)(\lambda - 2),$$

so

$$h_n^{(h)} = C + D2^n.$$

Because 2 is a root of the characteristic polynomial, use the resonant particular trial solution $h_n^{(p)} = E n 2^n$. Substitution gives

$$\begin{aligned} h_n^{(p)} - 3h_{n-1}^{(p)} + 2h_{n-2}^{(p)} &= E2^{n-2}(4n - 6(n-1) + 2(n-2)) \\ &= E2^{n-1}. \end{aligned}$$

Equating this with 2^n gives $E = 2$. Therefore

$$h_n = C + D2^n + 2n2^n.$$

The initial conditions imply

$$C + D = 0, \quad C + 2D + 4 = 1.$$

Thus $D = -3$ and $C = 3$, and

$$\boxed{h_n = 3 + (2n - 3)2^n \quad (n \geq 0).}$$

The formula gives $h_0 = 0$ and $h_1 = 1$, and the substitution above shows that it satisfies the recurrence.

5. Paths bounded below by a horizontal line

Write $U = (1, 1)$ and $D = (1, -1)$. Let A_r denote the number of paths from $(0, 0)$ to $(2n, 0)$ that stay in $y \geq -r$.

(a) Paths staying in $y \geq -r$

There are $\binom{2n}{n}$ unrestricted paths, since a path ending at height 0 has n up-steps and n down-steps. A path that violates the condition $y \geq -r$ must first hit the line $y = -(r+1)$. Reflect the portion of the path from its beginning through this first hitting point across the line $y = -(r+1)$. The starting point $(0, 0)$ is sent to $(0, -2r-2)$, while the endpoint remains $(2n, 0)$.

This reflection is a bijection between bad paths and unrestricted paths from $(0, -2r-2)$ to $(2n, 0)$. Such a path has $n+r+1$ up-steps and $n-r-1$ down-steps, so there are $\binom{2n}{n+r+1}$ of them. Consequently,

$$A_r = \binom{2n}{n} - \binom{2n}{n+r+1}.$$

The stated binomial convention makes this valid also when $r = n$.

(b) Paths with minimum height exactly $-r$

For $r \geq 1$, the paths staying above $-r$ but not above $-(r-1)$ are counted by $A_r - A_{r-1}$. Therefore

$$\begin{aligned} A_r - A_{r-1} &= \left[\binom{2n}{n} - \binom{2n}{n+r+1} \right] - \left[\binom{2n}{n} - \binom{2n}{n+r} \right] \\ &= \binom{2n}{n+r} - \binom{2n}{n+r+1}. \end{aligned}$$

When $r = 0$, every path with minimum height exactly 0 is simply a path staying in $y \geq 0$, and the same expression is A_0 . Hence for every $0 \leq r \leq n$,

$$\#\{\text{paths with minimum height exactly } -r\} = \binom{2n}{n+r} - \binom{2n}{n+r+1}.$$

6. Triangulations and Catalan numbers

(a) The recurrence

Consider the triangle incident to the boundary edge v_0v_{n+1} . Its third vertex is v_{k+1} for a unique k with $0 \leq k \leq n-1$. The diagonal sides of this triangle split the polygon into two independent polygons:

$$v_0, v_1, \dots, v_{k+1} \quad \text{and} \quad v_{k+1}, v_{k+2}, \dots, v_{n+1}.$$

The first has $k+2$ vertices and hence t_k triangulations; the second has $n-k+1$ vertices and hence t_{n-k-1} triangulations. Conversely, choosing triangulations of these two subpolygons and adjoining the distinguished triangle gives a unique triangulation of the original polygon. Thus

$$t_0 = 1, \quad t_n = \sum_{k=0}^{n-1} t_k t_{n-1-k} \quad (n \geq 1).$$

The Catalan numbers are characterized by the same recurrence

$$C_0 = 1, \quad C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}.$$

Induction on n therefore gives $t_n = C_n$ for every n . Equivalently, the generating function $C(z) = \sum_{n \geq 0} C_n z^n$ satisfies $C(z) = 1 + zC(z)^2$, whose solution with constant term 1 is

$$C(z) = \frac{1 - \sqrt{1 - 4z}}{2z} = \sum_{n \geq 0} \frac{1}{n+1} \binom{2n}{n} z^n.$$

Thus $t_n = C_n = \frac{1}{n+1} \binom{2n}{n}$.

(b) A prescribed diagonal

For $1 \leq r \leq n-1$, the diagonal $v_0 v_{r+1}$ splits the polygon into the subpolygons

$$v_0, v_1, \dots, v_{r+1} \quad \text{and} \quad v_0, v_{r+1}, v_{r+2}, \dots, v_{n+1}.$$

Their triangulations can be chosen independently. They have $r+2$ and $n-r+2$ vertices, respectively, so the number of triangulations containing the diagonal is

$$\boxed{t_r t_{n-r} = C_r C_{n-r}.$$

7. Fully bracketed expressions and ballot sequences

(a) The first construction is not injective

The two distinct bracketings

$$((a_1 a_2) a_3) \quad \text{and} \quad (a_1 (a_2 a_3))$$

both produce the word $++--$: in either case there are two opening parentheses followed, after deleting the letters, by two closing parentheses. Hence the construction in part (a) is not injective.

(b) The second construction gives a ballot sequence

There are n opening parentheses and n closing parentheses in a bracketing on $n+1$ letters. After deleting all closing parentheses and the final letter a_{n+1} , the construction therefore produces a word of length $2n$, containing n plus signs and n minus signs.

To check the prefix condition, view a bracketing as an ordered full binary tree: each opening parenthesis is an internal node and each letter is a leaf. Reading the expression from left to right is preorder traversal. Start with one available child slot. Reading an opening parenthesis fills one slot and creates two, so the number of available slots increases by one. Reading a letter fills one slot, so the number decreases by one. Before the final leaf a_{n+1} is read, at least one slot remains. Thus every prefix of the constructed word satisfies

$$1 + \#(+) - \#(-) \geq 1,$$

or equivalently $\#(+) \geq \#(-)$. The word is therefore a ballot sequence.

(c) The inverse and the enumeration

Let w be a ballot sequence of length $2n$ with n plus signs and n minus signs. Append one additional minus sign, which will represent the final letter a_{n+1} . Reconstruct an ordered full binary tree by processing this extended word from left to right:

- on a $+$, fill the leftmost available slot with an internal node and create its two ordered child slots;
- on a $-$, fill the leftmost available slot with a leaf.

After any prefix of w , the number of slots is $1 + \#(+)-\#(-) \geq 1$; after the final appended minus it is zero. Thus the algorithm is well-defined and produces a full binary tree with n internal nodes and $n+1$ leaves. Label the leaves from left to right by $a_1, \dots, a_{n+1}$, and write an internal node with left and right subexpressions U, V as (UV) . This yields a fully bracketed expression.

For an expression, the same slot procedure reads its preorder tree and recovers exactly the original tree. Conversely, the reconstructed tree has the prescribed preorder word, so deleting closing parentheses and the final leaf returns w . The two constructions are therefore inverse bijections.

The number of ballot sequences of length $2n$ is, by the reflection principle,

$$\binom{2n}{n} - \binom{2n}{n+1} = \frac{1}{n+1} \binom{2n}{n} = C_n.$$

Consequently, the number of fully bracketed expressions on $n+1$ letters is

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

8. Dyck paths counted by peaks

(a) The first three polynomials

For semilength 1, the only path is UD, with one peak:

$$P_1(q) = q.$$

For semilength 2, the paths are UUDD and UDUD, with one and two peaks:

$$P_2(q) = q + q^2.$$

For semilength 3, the five paths have peak counts 1, 2, 2, 2, 3, respectively. Thus

$$P_3(q) = q + 3q^2 + q^3.$$

(b) A first-return recurrence

Take a Dyck path D of semilength $n + 1$, and write its unique first-return decomposition as

$$D = \text{UADB},$$

where A is a Dyck path of semilength k and B is a Dyck path of semilength $n - k$, for some $0 \leq k \leq n$.

If $k = 0$, the initial UD is a peak and contributes a factor q ; the remaining path B contributes $P_n(q)$. This case contributes $qP_n(q)$. If $k \geq 1$, then A begins with an up-step, so the initial U is not immediately followed by a down-step. The peaks are exactly those in A and B , giving $P_k(q)P_{n-k}(q)$. Summing over all possibilities,

$$P_{n+1}(q) = qP_n(q) + \sum_{k=1}^n P_k(q)P_{n-k}(q) \quad (n \geq 0).$$

(c) Peaks and leaves in an ordered rooted tree

For an ordered rooted tree with $n + 1$ vertices, traverse each edge away from the root and record an up-step when an edge is traversed downward and a down-step when that edge is traversed upward. Each of the n edges is traversed twice, so this gives a Dyck path of length $2n$.

An up-step is immediately followed by a down-step exactly when the traversal reaches a vertex and has no child to visit before returning. This occurs exactly at a leaf. Conversely, every nonroot leaf creates such a peak. Since $n \geq 1$, the root is not a leaf, so

$$\text{the number of peaks equals the number of leaves of the tree.}$$

9. 132-avoiding permutations and Rothe diagrams

For a permutation σ , write

$$\text{pos}(j) = \sigma^{-1}(j).$$

Thus a cell in row i and column j of the Rothe diagram is present exactly when

$$j < \sigma(i) \quad \text{and} \quad \text{pos}(j) > i.$$

Let d_i be the number of cells in row i .

(a) Characterization by row lengths

132-avoidance implies left-justified rows. Suppose σ is 132-avoiding, and suppose row i contains a cell in column j . If some $\ell < j$ were not in row i , then $\ell < j < \sigma(i)$, and $\text{pos}(\ell) < i$, while $\text{pos}(j) > i$. Set $h = \text{pos}(\ell)$ and $k = \text{pos}(j)$. Then

$$h < i < k, \quad \sigma(h) = \ell < j = \sigma(k) < \sigma(i),$$

which is a 132 pattern. Therefore every row consists of its leftmost d_i cells.

132-avoidance implies weakly decreasing row lengths. Assume the rows are left-justified. If $d_{i+1} > d_i$, let $j = d_i + 1$. Then $(i + 1, j)$ is a cell, so

$$j < \sigma(i + 1), \quad \text{pos}(j) > i + 1.$$

The cell (i, j) is absent, and $\text{pos}(j) > i$, so $j \geq \sigma(i)$. Equality would force $\text{pos}(j) = i$, impossible; hence

$$\sigma(i) < j < \sigma(i + 1).$$

Using $k = \text{pos}(j)$, we have

$$i < i + 1 < k, \quad \sigma(i) < j = \sigma(k) < \sigma(i + 1),$$

again a 132 pattern. Hence $d_i \geq d_{i+1}$ for every i .

The converse. Now suppose every row is left-justified and

$$d_1 \geq d_2 \geq \cdots \geq d_n \geq 0.$$

If $i < j < k$ formed a 132 pattern, put $c = \sigma(k)$. Since $c < \sigma(j)$ and $\text{pos}(c) = k > j$, the cell (j, c) lies in row j . Left-justification gives $d_j \geq c$, and weak decrease gives $d_i \geq d_j \geq c$. Thus (i, c) would be a cell, which would imply $c < \sigma(i)$, contradicting the assumed inequality $\sigma(i) < c$. Therefore σ is 132-avoiding.

The bound on row lengths. Every cell in row i has its column value in a position strictly below i . There are only $n - i$ such positions, and distinct cells have distinct column values. Consequently

$$\boxed{d_i \leq n - i \quad (1 \leq i \leq n).}$$

(b) A bijection with Dyck paths

The characterization in part (a) shows that a 132-avoiding permutation is encoded by a sequence

$$d_1 \geq d_2 \geq \cdots \geq d_n \geq 0, \quad d_i \leq n - i.$$

We first explain how to recover the permutation from such a sequence. Process the rows from $i = 1$ to n . At step i , let $\sigma(i)$ be the smallest unused value that is strictly larger than d_i :

$$\boxed{\sigma(i) = \min\{j > d_i : j \text{ has not been used in rows } 1, \dots, i - 1\}.}$$

This set is nonempty. Indeed, all values at most d_i are still unused, because earlier rows have lengths at least d_i and therefore chose values larger than d_i . The number of unused values larger than d_i is at least

$$n - d_i - (i - 1) \geq 1$$

by $d_i \leq n - i$.

The values $1, \dots, d_i$ are then placed below row i , so they give exactly d_i cells. Any value j with $d_i < j < \sigma(i)$ was already used, by the minimality in the displayed rule, so it is not below row i . Thus the resulting permutation has precisely the prescribed Rothe diagram. The same minimality

condition shows that no other permutation can have these row lengths, so the reconstruction is unique.

It remains to convert the row-length sequence into a Dyck path. Define

$$e_i = d_{n-i+1} \quad (1 \leq i \leq n).$$

Then

$$0 \leq e_1 \leq e_2 \leq \cdots \leq e_n, \quad e_i \leq i - 1.$$

Construct the word

$$W(d) = D^{e_1} U D^{e_2 - e_1} U \cdots D^{e_n - e_{n-1}} U D^{n - e_n}.$$

Here $U = (1, 1)$ and $D = (1, -1)$, and a zero exponent means that the corresponding run is empty. The word has n up-steps and n down-steps. Before the i -th up-step there are $e_i \leq i - 1$ down-steps, so no prefix goes below height 0. Hence $W(d)$ is a Dyck path of length $2n$.

Conversely, given a Dyck path, let e_i be the number of down-steps before its i -th up-step. The Dyck condition gives $e_i \leq i - 1$, and the sequence e_i is weakly increasing. Set

$$d_i = e_{n-i+1}.$$

Then $d_1 \geq \cdots \geq d_n \geq 0$ and $d_i \leq n - i$. Reconstructing σ by the smallest-unused-value rule above gives a unique 132-avoiding permutation. These two procedures are inverse, so they give an explicit bijection between 132-avoiding permutations in S_n and Dyck paths of length $2n$. Therefore

$$\#\{132\text{-avoiding permutations in } S_n\} = C_n = \frac{1}{n+1} \binom{2n}{n}.$$

10. Monomials in a product of partial sums

$$Q_n(x_1, \dots, x_n) = \prod_{i=1}^n (x_1 + \cdots + x_i).$$

For a monomial $x_1^{b_1} \cdots x_n^{b_n}$, necessarily $\sum_{i=1}^n b_i = n$, because one variable is chosen from each of the n factors.

(a) The prefix condition

Suppose the monomial occurs. In the first j factors, every selected variable has index at most j . Those j selections therefore contribute to the total exponent of one of $x_1, \dots, x_j$. Consequently,

$$b_1 + \cdots + b_j \geq j \quad (1 \leq j \leq n).$$

Conversely, assume $\sum_{i=1}^n b_i = n$ and all these prefix inequalities hold. Assign a variable to each factor greedily from $i = 1$ through n , using exactly b_j assignments of x_j . At step i , there must be an unfilled demand among $x_1, \dots, x_i$. Otherwise all demands in this set would already have been filled by the first $i - 1$ factors, which would imply $b_1 + \cdots + b_i \leq i - 1$, contradicting the prefix condition. Choose any such x_j with $j \leq i$ from the i -th factor. After all n steps, every demand is filled, so the monomial occurs in Q_n . Thus

$$x_1^{b_1} \cdots x_n^{b_n} \text{ occurs in } Q_n \iff \sum_{i=1}^n b_i = n \text{ and } \sum_{i=1}^j b_i \geq j \text{ } (1 \leq j \leq n).$$

(b) Fixing the exponent of x_1

Encode an exponent vector satisfying part (a) by the word

$$W(b) = U^{b_1}D U^{b_2}D \dots U^{b_n}D.$$

There are n up-steps in total and n down-steps. After the j -th down-step, the height is

$$(b_1 + \dots + b_j) - j,$$

so the prefix inequalities in part (a) are exactly the Dyck-path condition. If $b_1 = r$, the path begins with exactly r up-steps followed by a down-step. Thus we count Dyck paths of semilength n whose initial run of up-steps has length exactly r .

After the forced prefix U^rD , the remaining path has length $2n - r - 1$, starts at height $r - 1$, ends at height 0, and contains $n - r$ up-steps and $n - 1$ down-steps. Ignoring the nonnegativity condition, there are $\binom{2n-r-1}{n-r}$ such paths. Reflecting the initial segment through the first visit to $y = -1$ is a bijection from the bad paths to paths starting at height $-r - 1$ and ending at 0. A reflected path has n up-steps, so the number of bad paths is $\binom{2n-r-1}{n}$. Therefore the number of exponent vectors, and hence of distinct monomials, is

$\#\{\text{monomials with exponent of } x_1 \text{ equal to } r\} = \binom{2n-r-1}{n-r} - \binom{2n-r-1}{n}, \quad 1 \leq r \leq n.$
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